

National University of Computer and Emerging Sciences



Lab Manual 05 Artificial Intelligence

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Lab Manual 5

Objectives:

- ✓ File Handling
csv & json files, Exception Handling
- ✓ OOP in Python
Polymorphism, Encapsulation, Abstraction, Modules & Packages
- ✓ Informed Search Algorithms
Greedy Best-First Search (GBFS), A* Search Algorithm, Minimax Algorithm

Activities:

Activity 1 (Python):

File Handling

Writing to a csv file

```
import csv

data = [{"Name", "Age"}, ["Alice", 30], ["Bob", 25]]
with open("data.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerows(data)
```

Reading from a csv file

```
import csv

with open("data.csv", "r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

Writing to a json file

```
import json

data = {"Name": "Alice", "Age": 30}
with open("data.json", "w") as file:
    json.dump(data, file, indent=4)

data_json = [{ 'name': 'Ali', 'age': 30, 'student': 'yes'},
               { 'name': 'Ahmad', 'age': 40, 'student': 'no'}]
with open('output.json', 'w') as file:
    json.dump(data_json, file, indent=4)
```

Reading from a json file

```
import json
```

```
with open("data.json", "r") as file:
    data = json.load(file)
    print(data)
```

Working on csv file

Reading

```
import csv
```

```
with open("seeds.csv", "r") as file:
    reader = csv.reader(file)
```

```
for r in reader:
    print(r)
```

Deleting the Header Row having column names

```
import csv
```

```
with open("seeds.csv", "r") as file:
    reader = csv.reader(file)
```

```
my_data = []
for r in reader:
    my_data.append(r)
```

```
del my_data[0]
print(my_data)
```

sum of each row

```
import csv
```

```
with open("seeds.csv", "r") as file:
    reader = csv.reader(file)
```

```
my_data = []
for r in reader:
    my_data.append(r)
```

```
del my_data[0]
print(my_data)
```

```
for row in my_data:
    sum_of_row = 0
    for element in row:
        sum_of_row += float(element)
    print(sum_of_row)
```

sum of each row with built-in sum function and list comprehension

```
import csv

with open("seeds.csv", "r") as file:
    reader = csv.reader(file)

    my_data = []
    for r in reader:
        my_data.append(r)

del my_data[0]
print(my_data)

for row in my_data:
    row = [float(item) for item in row]
    sum_of_row = sum(row)
    print(sum_of_row)
```

Loading Google Drive

```
from google.colab import drive
drive.mount('/content/drive')
```

Try and Except Blocks

```
# Execution stops due to error
# That's why we use try except blocks

# execution starts from 'try' block
# if 'try' executes successfully, 'else' also executes
# if 'try' block has an error, 'except' block executes
# 'finally' block always gets executed
```

Simple Exception

```
try:
    num = int(input("Enter a number: "))
    result = 10 / num
    print("Result:", result)
except ZeroDivisionError:
    print("Error: Division by zero is not allowed!")
except ValueError:
    print("Error: Invalid input. Please enter a valid number.")
```

Multiple Exception

```
try:
    num = int(input("Enter a number: "))
    result = 10 / num
    print("Result:", result)
```

```
except (ZeroDivisionError, ValueError) as e:
    print("An error occurred:", e)
```

else Block

```
try:
    num = int(input("Enter a number: "))
    result = 10 / num
    print("Result:", result)
except (ZeroDivisionError, ValueError) as e:
    print("An error occurred:", e)
else:
    print("Else Block")
```

finally Block

```
try:
    file = open("example.txt", "r")
    content = file.read()
    print(content)
except FileNotFoundError:
    print("Error: The file does not exist.")
finally:
    file.close()
    print("File closed successfully.")
```

'finally' without 'except'

```
try:
    print("trying to divide by zero.")
    result = 10 / 0
finally:
    print("This will always run, regardless of an error")
```

Raising Exception

```
def check_positive(num):
    if num < 0:
        raise ValueError("Negative numbers are not allowed.")
    return num

try:
    print(check_positive(5))
except ValueError as e:
    print("Error:", e)
```

Raising build-in exception

```
def divide(a, b):
    if b == 0:
        raise ZeroDivisionError("Cannot divide by zero!")
    return a / b
```

```

try:
    result = divide(10, 0)
    print("Result:", result)
except ZeroDivisionError as e:
    print("Error:", e)

```

Example:

```

def check_age(age):
    if age <= 0:
        raise ValueError('Age is either zero or negative')
    else:
        print('You have entered correct age.')

```

```

try:
    check_age(5)
except ValueError as e:
    print(e)

```

Custom Exceptions

Simple Custom Exception

```

class InvalidAgeError(Exception):
    """Exception raised for invalid age."""
    pass

```

```

def check_age(age):
    if age < 0:
        raise InvalidAgeError("Age cannot be negative!")
    print(f"Age is valid: {age}")

```

```

try:
    check_age(25)
    check_age(-5)
except InvalidAgeError as e:
    print("Error:", e)

```

Simple Custom Exception with super()

```

class WithdrawalError(Exception):
    """Raise Error if the withdraw amount is greater than balance"""
    def __init__(self, balance, amount):
        self.balance = balance
        self.amount = amount
        super().__init__(f"Insufficient balance! Tried to withdraw {self.amount}, but only {self.balance} is available")

def withdraw(balance, amount):
    if amount > balance:
        raise WithdrawalError(balance, amount)
    return balance - amount

```

```

try:
    print("New Balance:", withdraw(100, 50))
    print("New Balance:", withdraw(100, 150))
except WithdrawalError as e:
    print("Error:", e)

# Exercise 1: Count the number of words in a text file
with open("sample.txt", "r") as file:
    content = file.read()
    words = content.split()
    print(f'Number of words in the text file are: {len(words)}')

# Exercise 2: Read a CSV and find the average of a column
import csv

with open ('./sample_data/california_housing_test.csv') as file:
    reader = csv.DictReader(file)

    sum = 0
    count = 0

    for row in reader:
        sum += float(row['housing_median_age'])
        count += 1

    print(f'Average of the column "housing_median_age" is {sum/count}')

# Exercise 3: Implement code to handle errors during numerical input
class RangeError(Exception):
    """Raise Error if the input is out of range"""
    pass

def get_input():
    num = input('Please enter a number between 0 and 100: ')

    num = float(num)

    if num < 0 or num > 100:
        raise RangeError("Number is less than 0 or greater than 100")

    return num

while True:
    try:
        num = get_input()
        break
    except (RangeError, ValueError) as e:
        print(e)

```

Exercise 4: Implement a code to handle exception for invalid file formats - txt

```
class InvalidFileFormatError(Exception):
    """
    Exception to be raised if input file format is not a txt
    """

    def __init__(self, fn):
        self.filename = fn
        super().__init__(f'Invalid file format, only txt is supported')

try:
    path_file = "sample.txt"

    if not path_file.endswith('.txt'):
        raise InvalidFileFormatError(path_file)

    print('File format is correct.')
except InvalidFileFormatError as e:
    print(e)
```

OOP in Python

Advanced OOP Concepts - Polymorphism

```
class Animal:    # Parent Class
    def speak(self):
        return "Some generic animal sound"

class Dog (Animal):    # Child Class
    def speak(self):    # Overriding speak()
        return "Bark"

class Cat (Animal):    # Child Class
    def speak(self):    # Overriding speak()
        return "Meow"

# creating objects
dog = Dog()
cat = Cat()

print(dog.speak())
print(cat.speak())
# Example
# Base Class
class Animal:
    def __init__(self, name):
        self.name = name
        print(f"Animal Constructor Called for {self.name}")
```

```

def eat(self):
    print(f"{self.name} is eating.")

def speak(self):
    print('Generic Sound')

# Derived Class from Animal
class Dog(Animal):
    def __init__(self, name, breed):
        super().__init__(name) # Call parent constructor
        self.breed = breed
        print(f"Dog Constructor Called for {self.name}")

    def speak(self):
        print('Dog is barking.')

# Derived Class from Animal
class Fish(Animal):
    def __init__(self, name, water_type):
        Animal.__init__(self, name)
        self.water_type = water_type
        print(f"Fish Constructor Called for {self.name}")

    def swim(self):
        print(f"{self.name} is swimming in {self.water_type} water.")

    def speak(self):
        print('Fish are generally silent')

# Multilevel Inheritance (Dolphin -> Fish -> Animal)
class Dolphin(Fish):
    def __init__(self, name, water_type, intelligence_level):
        super().__init__(name, water_type)
        self.intelligence_level = intelligence_level
        print(f"Dolphin Constructor Called for {self.name}")

    def jump(self):
        print(f"{self.name} is jumping out of the water!")

    def speak(self):
        print('Dolphins are whistling')

# Creating Objects
print("----- Animal Object -----")
animal1 = Animal("Generic Animal")

print("\n----- Dog Object -----")
dog1 = Dog("Buddy", "Golden Retriever")

```

```

print("\n----- Fish Object -----")
fish1 = Fish("Nemo", "Fresh")

print("\n----- Dolphin Object -----")
dolphin1 = Dolphin("Flipper", "Salt", "High")

# Calling functions for each object
print("\n----- Animal Object Function -----")
animal1.speak()

print("\n----- Dog Object Function -----")
dog1.speak()

print("\n----- Fish Object Functions -----")
fish1.speak()

print("\n----- Dolphin Object Functions -----")
dolphin1.speak()

```

Advanced OOP Concepts - Encapsulation

```

# public_var
# _protected_var - single underscore is just the convention,
# it does not stop being access outside the classes
# __private_var

class BankAccount:
    def __init__(self, owner, balance):
        self.owner = owner          # public variable
        self.__balance = balance    # private variable

# Creating an object
account = BankAccount("Alice", 1000)

# Trying to access the private attribute
print(account.owner)               # works fine
print(account.__balance)           # AttributeError

```

Getters & Setters Functions

```

class BankAccount:
    def __init__(self, owner, balance):
        self.owner = owner          # public variable
        self.__balance = balance    # private variable

    def get_balance(self):           # Getter Method
        return self.__balance

```

```

def set_balance(self, amount):          # Setter Method
    if amount < 0:
        print("Balance cannot be negative!")
    else:
        self.__balance = amount

# Creating an object
account = BankAccount("Alice", 1000)

# Accessing balance using getter
print(account.get_balance())

# Updating balance using setter
account.set_balance(1500)

# Again accessing balance using getter
print(account.get_balance())

```

Example

```

class BankAccount:
    def __init__(self, name, bal):
        self.name = name
        self.__balance = bal

    def deposit(self, amt):
        self.__balance += amt

    def print_bal(self):
        print(self.__balance)      # Works fine

    def get_balance(self):
        return self.__balance

    def set_balance(self, amount):
        if self.__balance == 0:
            if amount < 0:
                print("Balance cannot be negative.")
            else:
                self.__balance += amount
        else:
            print("Balance already set.")

class SavingAccount(BankAccount):
    def __init__(self, name, bal):
        super().__init__(name, bal)
        self.acc_type = "Savings"

```

```

def print_bal(self):
    print(self.__balance)      # AttributeError

b1 = BankAccount('Ali', 500)
b1.print_bal()
b1.set_balance(100)
print(b1.get_balance())

b2 = SavingAccount('Ahmad', 500)
b2.print_bal()

```

Advanced OOP Concepts - Abstraction

```

from abc import ABC, abstractmethod
# Abstract Class
class Vehicle(ABC):
    @abstractmethod
    def start(self):
        pass    # No implementation, subclasses must define this method

    @abstractmethod
    def stop(self):
        pass    # No implementation, subclasses must define this method

# Concrete Class (inherits from abstract class)
class car(Vehicle):
    def start(self):
        print("Car is starting...")

    def stop(self):
        print("Car is stoping...")

my_car = car()    # concrete class can be instantiated while abstract
class can't
my_car.start()
my_car.stop()

```

Example

```

from abc import ABC, abstractmethod

# Abstract Class
class BankAccount(ABC):
    def __init__(self, name, balance):
        self.name = name
        self.balance = balance

    @abstractmethod
    def deposit(self, amount):
        pass    # No implementation, subclasses must define this method

```

```

@abstractmethod
def withdraw(self, amount):
    pass          # No implementation, subclasses must define this method

# Concrete Class
class SavingAccount(BankAccount):
    def deposit(self, amount):
        if amount < 0:
            print("Cannot deposit negative amount.")
        else:
            self.balance += amount
            print(f"Deposited ${amount}. New Balance: ${self.balance}")

    def withdraw(self, amount):
        if amount < 0:
            print("Cannot withdraw negative amount.")
        elif amount > self.balance:
            print("Insufficient Funds!")
        else:
            self.balance -= amount
            print(f"Withdrawn ${amount}. New Balance: ${self.balance}")

# Creating an object
account = SavingAccount('Ali', 1000)
account.deposit(500)
account.withdraw(200)
account.withdraw(2000)

```

Exercise: Implement a banking system with classes Bank, Customer and Account

```

class Account:
    def __init__(self, balance=0):
        self.__balance = balance

    def get_balance(self):
        return self.__balance

    def deposit(self, amt):
        self.__balance += amt

    def withdraw(self, amt):
        if self.__balance >= amt:
            self.__balance -= amt
            print(f'Remaining Balance: {self.__balance}')
        else:
            print('Insufficient funds')

```

```

class Customer:
    def __init__(self, id, name, add):
        self.__id = id
        self.name = name
        self.address = add
        self.account = Account()

    def get_name(self):
        return self.name

    def get_address(self):
        return self.address

    def set_address(self, add):
        self.address = add
        print(f'Address updated to {add}')

    def deposit(self, amt):
        self.account.deposit(amt)

    def withdraw(self, amt):
        self.account.withdraw(amt)

    def get_balance(self):
        return self.account.get_balance()

class Bank:
    def __init__(self, bn):
        self.bank_name = bn
        self.customers = {}

    def add_customer(self, id, n, add):
        self.customers[id] = Customer(id, n, add)

    def get_customer(self, id):
        return self.customers.get(id, 'None')

bank = Bank('ABC Bank')

bank.add_customer(1, 'Ben', 'XYZ Street')

current_customer = bank.get_customer(1)

current_customer.deposit(200)
print(current_customer.get_balance())

current_customer.deposit(100)
print(current_customer.get_balance())

```

```
current_customer.withdraw(200)
print(current_customer.get_balance())
```

```
current_customer.withdraw(150)
print(current_customer.get_balance())
```

Modules

#Example

```
%%writefile my_first_module.py
```

```
def add(a, b):
    return a + b
```

```
def subtract(a, b):
    return a - b
```

```
def frequency_unique(df_col):
    '''
    This function returns the frequency of unique values in a python dict format
    '''
    val_counts = df_col.value_counts().reset_index()

    output_dict = {float(row[1][0]): int(row[1][1]) for row in
val_counts.iterrows()}

    return output_dict
```

```
# importing my_first_module.py
```

```
import my_first_module
```

```
import pandas as pd
```

```
my_first_module.add(4, 5)
```

```
df = pd.read_csv('sample_data/california_housing_test.csv')
```

```
my_first_module.frequency_unique(df['housing_median_age'])
```

Packages

```
# Many modules in one package
```

```
!mkdir my_package
```

```
# Next Cell
```

```
%%writefile my_package/__init__.py
```

```
# Next Cell
```

```

%%writefile my_package/math_operations.py
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def frequency_unique(df_col):
    '''
    This function returns the frequency of unique values in a python dict
    format
    '''
    val_counts = df_col.value_counts().reset_index()

    output_dict = {float(row[1][0]): int(row[1][1])
                    for row in val_counts.iterrows()}

    return output_dict

# Next Cell
%%writefile my_package/greeting_operations.py

def greet(name):
    print(f'Hello {name}, welcome to code')

def bye(name):
    print(f'Good Bye {name}, We hope you enjoyed our program.')

# Importing Modules & Functions
from my_package import math_operations
result = math_operations.subtract(3, 1)
print(result)

from math import *
print(cos(0))

import pandas as pd
data = pd.DataFrame({"A": [1, 2, 3], "B": [4, 5, 6]})
print(data)

```

Activity 2 (Informed Search Algorithms):

Informed Search Algorithms

Informed search algorithms (also called heuristic search algorithms) use additional problem-specific knowledge to guide the search toward the goal more efficiently.

Unlike blind search algorithms (BFS, DFS, UCS), informed search algorithms use a heuristic function to estimate how close a state is to the goal.

They make use of:

- Initial state
- Goal state
- Actions available between states
- Heuristic information

What is a Heuristic?

A heuristic function, denoted as $h(n)$, estimates the cost from node n to the goal. It does not give the exact cost — it gives an estimate.

Properties of a Good Heuristic:

- Problem-specific
- Improves search efficiency
- Should not overestimate (for optimality in A*)

Heuristic Function in Python

A heuristic is typically implemented as a dictionary or a function:

```
heuristic = {
    'A': 6,
    'B': 4,
    'C': 2,
    'D': 3,
    'E': 1,
    'G': 0
}
```

or

```
def h(state):
    return heuristic[state]
```

Types of Informed Search Algorithms

Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the node that appears closest to the goal.

Evaluation Function:

$$f(n) = h(n)$$

It ignores the path cost and uses only heuristic value.

Algorithm

- Create priority queue ordered by $h(n)$
- Insert start node
- While queue not empty:
 - Remove node with smallest $h(n)$
 - If goal → return solution
 - Expand neighbors
 - Insert unexplored neighbors

Python Implementation

```
import heapq

def greedy_bfs(graph, heuristic, start, goal):
    pq = []
    heapq.heappush(pq, (heuristic[start], Node(start)))
    visited = set()

    while pq:
        current_h_value, current_node = heapq.heappop(pq)
        current_state = current_node.state

        if current_state == goal:
            return get_path(current_node)

        visited.add(current_state)

        for neighbor in graph[current_state]:
            if neighbor not in visited:
                child = Node(neighbor, current_node)
                heapq.heappush(pq, (heuristic[neighbor], child))

    return None
```

Properties

Property	Value
Optimal	No
Complete	Not always
Memory	Low
Speed	Fast

A* Search Algorithm

A* combines:

- Uniform Cost Search (path cost)
- Greedy Best-First Search (heuristic)

Evaluation Function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ → Cost from start to n
- $h(n)$ → Estimated cost from n to goal

Algorithm

- Insert start node with cost 0
- While queue not empty:
 - Remove node with smallest $f(n)$
 - If goal → return
 - Expand neighbors
 - Compute new $g(n)$, $h(n)$, $f(n)$
 - Update queue if better path found

Python Implementation

```
import heapq

def astar(graph, heuristic, start, goal):
    pq = []
    heapq.heappush(pq, (0, Node(start, cost=0)))
    visited = {}

    while pq:
        f_cost, current_node = heapq.heappop(pq)
        current_state = current_node.state

        if current_state == goal:
            return get_path(current_node)

        if current_state not in visited or f_cost < visited[current_state]:
            visited[current_state] = f_cost

            for neighbor, edge_cost in graph[current_state]:
                g = current_node.cost + edge_cost
                h_val = heuristic[neighbor]
                f = g + h_val

                child = Node(neighbor, current_node, g)
                heapq.heappush(pq, (f, child))

    return None
```

Properties

Property	Value
Optimal	No
Complete	Not always
Memory	Low
Speed	Fast

Comparison Summary

Algorithm	Evaluation	Optimal	Complete
GBFS	$h(n)$	No	No
A^*	$g(n) + h(n)$	Yes	Yes

Game Playing in Artificial Intelligence

Game playing is one of the classical problems in Artificial Intelligence. In game-playing problems, an intelligent agent must make decisions by considering possible future moves and the responses of an opponent.

These problems are usually modelled as:

- Two-player
- Turn-based
- Zero-sum games

Examples include:

- Tic-Tac-Toe
- Chess
- Checkers
- Connect Four

Characteristics of Game Playing Problems

- There are two players:
 - MAX (the AI or computer)
 - MIN (the opponent)
- Players take turns
- One player's gain is the other player's loss
- Both players are assumed to play optimally

Game State Representation

In game playing:

- A state represents a configuration of the game board
- The initial state is the starting position
- Terminal states are game-ending positions (win, lose, or draw)
- Each state has possible actions leading to new states
- The game is represented as a game tree.

Game Tree

A game tree is a tree structure where:

- Each node represents a game state
- Each edge represents a possible move
- Levels of the tree alternate between MAX and MIN players
- MAX nodes: try to maximize the score
- MIN nodes: try to minimize the score

Utility (Evaluation) Function

At terminal states, a utility function assigns a numeric value indicating the outcome of the game.

Typical utility values:

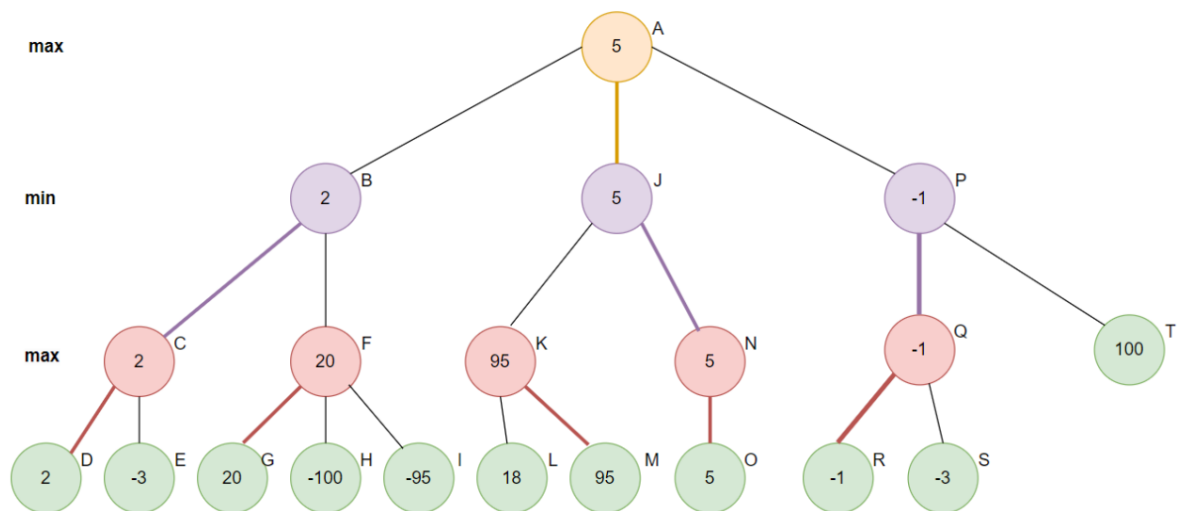
- Win for MAX: +1 (or positive value)
- Loss for MAX: -1 (or negative value)
- Draw: 0

Minimax Algorithm

The Minimax algorithm is a recursive decision-making algorithm used for optimal play in two-player games.

It assumes:

- Both players play optimally
- MAX tries to maximize the utility
- MIN tries to minimize the utility



Algorithm:

- Start from the root node
- If the node is a terminal state:
 - Return its utility value
- If it is MAX's turn:
 - Evaluate all child nodes
 - Return the maximum value
- If it is MIN's turn:
 - Evaluate all child nodes
 - Return the minimum value

Pseudocode:

MINIMAX(node, isMax):

```

if node is terminal:
    return utility(node)

if isMax:
    best =  $-\infty$ 
    for each child:
        value = MINIMAX(child, False)
        best = max(best, value)
    return best

else:
    best =  $+\infty$ 
    for each child:
        value = MINIMAX(child, True)
        best = min(best, value)
    return best
  
```

Node Representation:

A node in a game tree may contain:

- Game state
- List of possible moves
- Player type (MAX or MIN)
- Utility value (for terminal nodes)

For example:

```
class GameNode:
    def __init__(self, value=None, children=None):
        self.value = value
        self.children = children
```

Python Implementation of Minimax

```
def minimax(node, is_max):
    # Terminal node
    if not node.children:
        return node.value

    if is_max:
        best = float('-inf')
        for child in node.children:
            value = minimax(child, False)
            best = max(best, value)
        return best
    else:
        best = float('inf')
        for child in node.children:
            value = minimax(child, True)
            best = min(best, value)
        return best

leaf1 = GameNode(3)
leaf2 = GameNode(5)
leaf3 = GameNode(2)
leaf4 = GameNode(9)

min1 = GameNode(children=[leaf1, leaf2])
min2 = GameNode(children=[leaf3, leaf4])

root = GameNode(children=[min1, min2])

print("Optimal Value:", minimax(root, True))
```

Activity 3 (Problem Statements / Tasks):

1. *CSV Student Performance Analyzer*

You are given a CSV file named students.csv. Write a Python program that:

- Reads the CSV file.
- Remove the header row.
- For each student:
 - Convert marks to float.
 - Handle invalid values using try/except.
- Calculate:
 - Total marks
 - Average marks
- Display:
 - Student name
 - Total marks
 - Average marks
- Raise a custom exception and use try/except if:
 - Any student has negative marks.
 - File not found

2. *Inventory Management System*

Design a simple Inventory Management System using OOP and store product data in a JSON file.

Requirements

Part A – OOP Design

- Create a class Product with:
 - Private variable: __quantity
 - Instance variables:
 - name
 - price
 - Methods:
 - add_stock(amount)
 - sell_stock(amount)
 - get_quantity()
 - Rules:
 - Stock cannot go negative.
 - Price must be positive.
- Create a class Inventory
 - Instance variable:
 - products (dictionary storing Product objects)
 - Methods:
 - add_product(name, price, quantity)
 - sell_product(name, quantity)
 - restock_product(name, quantity)
 - display_inventory()

Part B – JSON Storage

Use a given file named: inventory_data.json. Program should:

- Load existing product data from JSON (if file exists).
- Convert JSON data into Product objects.

- Allow user to:
 - Add new product
 - Sell product
 - Restock product
 - View inventory
- Save updated data back to JSON file.

Error Handling

Your program must handle:

- File not found (create new file)
- Invalid numeric input
- Selling more stock than available
- Negative quantity
- Invalid product name

3. *Implement Greedy and A* Search*

Given the following weighted graph and heuristic values:

```
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [],
    'E': [],
    'F': []
}
```

```
heuristic = {
    'A': 7,
    'B': 6,
    'C': 2,
    'D': 1,
    'E': 3,
    'F': 0
}
```

Requirements

- Implement:
 - Greedy Best First Search
 - A* Search
- Find path from A to F.
- Display:
 - Path found by Greedy
 - Path found by A*
- Compare results and explain:
 - Which algorithm found better path?
 - Why?

4. *Minimax Tree Evaluation*

Construct the following game tree manually using GameNode class:

- Root → MAX
- Level 1 → MIN nodes
- Leaves contain values:
 - 3, 12
 - 8, 2
 - 4, 6

Requirements

- Create the tree structure.
- Apply Minimax algorithm.
- Display:
 - Value chosen at each MIN node
 - Final optimal value at root
- Explain:
 - Why MAX chose that value?